

Advanced (Habanero)

- Q1.** Which lines are parallel if they have slopes $m_1 = 2$ and $m_2 = 2$?
- (A) Lines with the same y -intercept
(B) Lines with the same slope
(C) Lines with different y -intercepts
(D) Lines with negative slopes
(E) Lines with positive y -intercepts
- Q2.** Given $y = 3x + 1$, what is the slope of a parallel line?
- (A) 0
(B) 1
(C) 2
(D) 3
(E) -3
- Q3.** Which equation represents a line parallel to $y = 2x - 3$?
- (A) $y = x - 2$
(B) $y = 2x + 1$
(C) $y = -2x + 3$
(D) $y = x + 4$
(E) $y = 3x - 1$
- Q4.** What is the relationship between the y -intercepts of parallel lines?
- (A) They are equal
(B) They are different
(C) They are negative
(D) They are positive
(E) They are zero
- Q5.** If two lines have slopes $m_1 = 4$ and $m_2 = -4$, are they parallel?
- (A) Yes
(B) No
(C) Maybe
(D) Only if they have the same y -intercept
(E) Only if they have different y -intercepts
- Q6.** Which of the following lines are parallel to $y = x - 2$?
- (A) $y = x + 1$ and $y = x - 3$
(B) $y = 2x + 1$ and $y = 2x - 2$
(C) $y = -x + 2$ and $y = x + 1$
(D) $y = x + 2$ and $y = x - 1$
(E) $y = x + 3$ and $y = x - 4$
- Q7.** What is the equation of a line parallel to $y = 2x + 1$ that passes through $(1, 3)$?
- (A) $y = 2x + 2$
(B) $y = 2x - 1$
(C) $y = x + 2$
(D) $y = x - 1$
(E) $y = 3x - 2$
- Q8.** If $y = mx + b$ and $y = nx + c$ are parallel, what is the relationship between m and n ?
- (A) $m = n$
(B) $m = -n$
(C) $m = 2n$
(D) $m = n + 1$
(E) $m = n - 2$

Open Ended Questions (FRQ)

- Q9.** Given two parallel lines $y = 2x + 1$ and $y = 2x - 3$, find the distance between them. Show your work.
- Q10.** Prove that if two lines have the same slope, they are parallel. Use the slope-intercept form of a line $y = mx + b$.

Chef's Tips

- Emphasize the importance of understanding the concept of parallel lines
- Provide examples of real-world applications of parallel lines
- Encourage students to use graphs and tables to visualize parallel lines